

Jaymin Sheladia

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EDUCATION

University of Southern California

Los Angeles, USA

Master of Science in Computer Science

January 2026 – December 2027

- **Coursework:** Analysis of Algorithms, Agentic AI, Applied NLP, Web Technologies, Database Systems

Graphic Era Deemed to be University

Dehradun, India

B.Tech in Computer Science and Engineering

October 2021 – June 2025

- **Coursework:** Operating Systems, Network & System Security, Computer Networks, OOPs, Software Engineering

TECHNICAL SKILLS

Languages: C++, Python, Java, C, Go, JavaScript/TypeScript, SQL, HTML/CSS

Systems & Databases: PostgreSQL, MongoDB, Redis, MySQL, Database Internals, Query Optimization, Linux

Frameworks & Tools: React, Node.js, FastAPI, REST APIs, Express, Flask, Next.js, Angular, GraphQL, GDB, GCC

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, Firebase, Git, CI/CD, Microservices

AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, RAG, SpaCy, YOLOv11, Claude API, Voyage AI

WORK EXPERIENCE

Cognizant Technology Solutions

May 2025 – September 2025

Programmer Analyst Trainee – Software Engineering

Coimbatore, India

- Engineered a full-stack Travel Planner and Booking System using Spring Boot and Angular, designing and integrating **10+** RESTful APIs to centralize itinerary, booking, and user preference data across the booking workflow.
- Integrated a Generative AI-based recommendation engine that evaluates **3** key signals – user interests, budget, and trip duration – to generate context-aware personalized travel recommendations for users.

CODTECH IT SOLUTIONS

June 2024 – August 2024

Machine Learning Intern

Remote

- Designed and trained a fraud detection model using anomaly detection and supervised machine learning on over **200,000** credit card transactions, reaching **98%** classification accuracy on live transaction data.
- Optimized machine learning algorithms and re-engineered data preprocessing pipelines for production models, improving predictive accuracy by **12–15%** across multiple live projects and reporting dashboards.

PROJECTS

Model Router | GitHub | Python, FastAPI, SQLite

July 2026 – August 2026

- Architected a policy-aware LLM routing service layering identity verification, per-project authorization, and budget enforcement across Groq and Claude models, with every decision logged to a SQLite audit trail.
- Benchmarked the routing classifier against a hand-labeled oracle set using live model inference, reaching **100%** classification accuracy while cutting inference cost by **27.9%** versus always calling the top-tier model.

TechHire | GitHub | Python, FastAPI, React, PostgreSQL, Redis

June 2026 – August 2026

- Developed a full-stack AI-powered job search platform using React, FastAPI, PostgreSQL, and Redis, aggregating **1,037** real job postings from **27+** company boards via Greenhouse, Lever, and Ashby integrations.
- Devised an AI-driven job intelligence pipeline using **3 Groq LLMs** run concurrently with a synthesis pass, generating context-aware job summaries, resume-to-job scoring, and section-level rewrite recommendations.
- Streamlined Redis-backed query caching and rate limiting, reducing mean request latency from **5.81 ms** to **2.81 ms** (**2.1x** speedup) while sustaining approximately **498 req/s** at **40** concurrent requests.

AI Research Assistant | GitHub | Python, Next.js, Gemini API

June 2026 – July 2026

- Built a full-stack AI research assistant with a PostgreSQL (pgvector) backend and Next.js frontend, implementing a RAG pipeline for semantic retrieval, contextual Q&A, and citation knowledge-graph traversal across ingested papers.
- Profiled and benchmarked the end-to-end document ingestion pipeline across live LLM inference, embedding, and persistence stages, achieving latency below **4.3 seconds** per paper, with LLM inference as **85–95%** of total latency.

Sequence Alignment: DP vs. D&C | GitHub | Python, Algorithms

March 2026 – April 2026

- Implemented and comparatively evaluated classical Needleman-Wunsch Dynamic Programming against Hirschberg's divide-and-conquer algorithm for global DNA sequence alignment, preserving optimal alignment correctness.
- Reduced memory usage from **$O(m \times n)$** to **$O(m + n)$** using Hirschberg's algorithm – a reduction of up to **90%** on large inputs – while preserving identical optimal alignment results.

RESEARCH WORK & PUBLICATIONS

Deep Learning Approach for Malicious URL Detection | Co-author; Published Research Paper 2025

- Compared **4** deep learning architectures (CNN, RNN, LSTM, Bi-LSTM) for malicious URL detection, engineering character-level preprocessing pipelines on the IEEE Dataport dataset for sequential analysis at scale.
- Achieved **99.25%** accuracy with the Bi-LSTM model, outperforming CNN (**86.5%**), LSTM (**63.75%**), and RNN (**53%**) baselines across standard classification and error-rate metrics.